Captain’s Summer Exchange - Alabama School of Fine Arts - Repaired practice edition - Answer key

Division C | Chemistry Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

16. (2 points) Authored replacement for source item Question 16: Overall coefficients always give rate orders?

Unit: Kinetics, rate laws, mechanisms, and activation energy.
Worked-case data: A first-order reaction has k = 0.0200 s□^1. Find half-life and fraction remaining after two half-lives.

A. Yes for every reaction

B. Only when products are gases

C. Only at equilibrium

D. No

Correct: D

18. (2 points) Authored replacement for source item Question 18: A double bond counts as how many VSEPR domains?

Unit: Bonding, molecular geometry, and intermolecular forces.
Worked-case data: Why is CO_2 nonpolar while H_2O is polar?

A. One

B. Two

C. Three

D. Zero

Correct: A

19. (4 points) Authored replacement for source item Question 19: Convert 0.20 mol CO_2 to moles of oxygen atoms.

answer: 0.40 mol oxygen atoms, because each molecule contains two. (4 points).